

Chapter 2

Analytical Framework

2.1 Introduction

To ensure an analysis has some foreshadowing, it is important to present the analytical frameworks to be compared beforehand. The pertinent frameworks are drawn from chapters III and IV of Vincent d'Indy's *Cours de composition musicale*: Book 2 - Part 1 [d'Indy, 1909]. For Caplin's approach, the framework is derived from his *Classical Form* [Caplin, 1998] and *Analyzing Classical Form* [Caplin, 2013], as well as classic articles such as "Expanded Cadential Progression" [Caplin, 1987].



2.2 Indy's 3 volumes

Vincent d'Indy authored three volumes of his *Cours de Composition Musicale*, a significant work in music theory. The first volume was published in 1903 by A. Durand et Fils in Paris. This initial publication laid the foundation for the subsequent parts of the series. The second volume was released in two parts: the first part was published in 1909 and the second part in 1933, written with the collaboration of Auguste Sérieyx based on notes taken from composition classes at the Schola Cantorum. The third and final volume was edited by Guy de Lioncourt and published in 1950 by Durand et Co. Rémy Campos highlights how the sequence of topics in the books follows the unfolding of the 3-year curriculum at the Schola

Cantorum:

In the first year, students study "the purely rhythmic and melodic forms of Gregorian Monody and Folk Song, then the polyphonic forms of the Motet and Madrigal, stopping at the beginning of the 17th century." The following year, the fugue is taught not "as a writing exercise but as a form of musical composition"; students also study the instrumental suite and "the Sonata in its three manifestations (the early Italian sonata, the Beethovenian sonata, and the modern cyclic form sonata)." In the third year, the master addresses the art of orchestration as well as the "forms of the Concerto and Concert, Symphony, Chamber Music, String Quartet, Variation, Fantasy, Overture, and Symphonic Poem." Finally, the last two years are dedicated to musical drama, from the birth of opera in the 17th century to the most modern works. [Campos, 2013, pp. 72-3, *translation mine*]

The first volume covers both fundamental musical parameters and a chronological survey of vocal and instrumental genres. On one hand, it addresses foundational elements such as rhythm, melody, notation, harmony, tonality, and expression, providing a solid theoretical basis for musical understanding and practice. On the other hand, it traces the historical development of vocal genres—including monophonic chant, motets, songs, and madrigals—from the Middle Ages through the Renaissance, situating these forms within the progressive evolution of musical art. This comprehensive approach reflects d'Indy's desire to establish a solid theoretical foundation for composers.

The second volume, published in two parts, delves into later instrumental genres. The first part addresses the fugue, the suite, pre-Beethoven sonatas, Beethoven's sonatas, the cyclic sonata, and variations. The second part focuses on larger-scale forms, discussing the concerto, the symphony, chamber music with accompaniment, the string quartet, the symphonic overture, the symphonic poem, and the fantasia.

The third volume is dedicated to dramatic music, with chapters on oratorio,

Chapter	Content
Chapter 1	Rythme
Chapter 2	Mélodie
Chapter 3	Notation
Chapter 4	La Cantilène Monodique
Chapter 5	La Chanson Populaire
Chapter 6	L'Harmonie
Chapter 7	La Tonalité
Chapter 8	L'Expression
Chapter 9	Histoire des Théories Harmoniques
Chapter 10	Le Motet
Chapter 11	La Chanson et le Madrigal
Chapter 12	L'Évolution Progressive de l'Art

Table 2.1: Book 1 Chapters

Chapter	Content
Chapter 1	Fugue
Chapter 2	La Suite
Chapter 3	Sonate Pré-Beethovénienne
Chapter 4	La Sonate de Beethoven
Chapter 5	La Sonate Cyclique
Chapter 6	Variation

Table 2.2: Book 2 - Part 1 Chapters

Chapter	Content
Chapter 1	Le Concert et le Concerto à Soliste
Chapter 2	La Symphonie Proprement Dite
Chapter 3	La Musique de Chambre Accompagnée
Chapter 4	Le Quatuor à Cordes Seules
Chapter 5	L'Ouverture Symphonique
Chapter 6	Le Poème Symphonique et la Fantaisie

Table 2.3: Book 2 - Part 2 Chapters

cantata, incidental music, and the lied. This final volume marks a shift from the instrumental forms explored in the earlier volumes to vocal and stage-based genres.

Chapter	Content
Chapter 1	La Musique Dramatique
Chapter 2	L'Oratorio et la Cantate
Chapter 3	La Musique de Scène
Chapter 4	Le Lied

Table 2.4: Book 3 Chapters

Each chapter of these three books is divided into two parts: "Technique" and "Historique." The "Technique" section focuses on the structural elements of musical genres and forms, while the "Historique" section looks at their historical development. For example, in Chapter 2 of book II, the "Technique" section addresses the definitions, origins of the suite (such as the transcription of songs for instruments and the modulating binary form), and the analysis of various movements (slow, moderate, and rapid). The "Historique" section then examines the historical development of the suite, focusing on early precursors and the contributions of Italian, French, and German composers.

The primary focus on the theory of form can be found in Chapter 3 of Volume II, where d'Indy outlines essential aspects of musical structure. However, the larger structural considerations relevant to Beethoven's music are interspersed throughout

the volumes, making it necessary to examine these sections in the broader context of his writings. For instance, the concept of sonata form in the current chapter builds on the foundation laid in the previous chapter, where d'Indy explains the pre-Beethovenian sonata form. Additionally, a walkthrough of Beethoven's symphonic work is found in Chapter 2 of Part II of the same volume, providing further opportunities to apply and detail d'Indy's theory.

2.3 Beethovenian Sonata Form

2.3.1 Overview of Pre-Beethovenian Sonata Form

In chapter III of Book 2 - Part 2 Pre-Beethovenian Sonata, d'Indy categorizes pre-beethovenian sonatas into two types: the monothematic sonata, containing one theme, and the dithematic sonata form, containing two themes.

While d'Indy dedicates a single chapter to Beethovenian sonatas later, his broader perspective on the structure of Beethovenian sonatas can be seen as an expansion of the architecture of the dithematic sonata form presented in the Pre-Beethovenian Sonata Chapter.

It is sufficient to acknowledge this fact without presenting the complete analytical framework for the pre-Beethovenian dithematic sonata form, as its architecture generally aligns with common sense or textbook standards of sonata form. According to this nomenclature, however, the classification of two themes suggests that d'Indy emphasizes a duality of themes rather than contrasting tonal regions within the exposition.

2.3.2 Beethovenian Sonata Form

Since the architecture of the Beethovenian sonata form is inherited from the skeleton of the dithematic sonata form introduced in the previous chapter, this chapter focuses on the generation of musical ideas and the techniques used to develop these ideas, with a particular emphasis on modulation techniques.

2.3.3 Chapter 3 of Volume II "Beethovenian Sonata" - An Overview

In the first half, "Technique" section [d'Indy, 1909, pp. 232-323], begins with a discussion of d'indyst notions such as *L'idée musicale*, *Développement*¹, *Modulation*, and the four main movement types found in the sonata form according to him.

The discussion of the *Idée musicale* focuses on addressing the structure and development of musical ideas. D'Indy introduces the cell as the smallest indivisible unit (rhythmic, melodic, or harmonic), which combines into periods and phrases to express a full musical idea. D'Indy emphasizes that this hierarchy—cell, period, phrase—is central to the work of composers like Beethoven, Wagner, and Franck. He also accentuates on the generative period, which contains the primary theme and acts as the foundation for the development of other periods and phrases. This process, central to Beethoven's sonata form, allows the musical idea to evolve while maintaining thematic unity. D'Indy uses examples such as the beginning measures of Beethoven's Piano Sonata Op. 106 and Symphony no. 5, Op. 67 to illustrate these concepts.

In discussing the notions of *Développement* and *Modulation*, d'Indy identifies two types of developments: organic and tonal. Organic development manipulates rhythmic, melodic, or harmonic elements, while tonal development focuses on the stability and modulation of keys. He highlights three types of modulation (accidental, passing, definitive) and their impact on the intensity and emotional direction of a piece.

The section then outlines the specific movement types found in sonata form: the *Mouvement Initial* (Type S, corresponding to a movement of sonata form), the *Mouvement Lent* (Type L, where the tempo is slow), the *Mouvement Modéré* (Type M, which includes the Menuet and Scherzo), and the *Mouvement Rapide* (Type R, such as the Rondo). In *Mouvement Initial*, d'Indy introduced a comprehensive

¹In this context, *Développement* does not designate a formal section, but the technique of developing the musical idea, such as transform it over the course of a movement.

walk through of the first movement of Beethoven's Piano Sonata Op. 106, illustrating how the structural units—cell, period, and phrase—function within a complete movement.

The second half, "Historique" section, provides a chronological examination of Beethoven's sonatas. It starts with the Chronology of Beethoven's Sonatas, giving context to the evolution of his style over time. This is followed by an analysis of Beethoven's Piano Sonatas, categorized into three distinct time periods: the **First time period** up to 1800, the **Second time period** up to 1815, and the **Third time period** from 1815 to 1826, in which each of Beethoven's 32 sonatas is briefly analyzed.² While some are only presented with an overview of the movement types and a basic description of their structural organization, the majority include more detailed illustrations of specific themes and their development in the movement. The chapter concludes by briefly mentioning Beethoven's Sonatas for Violin, for Cello, and for other instruments, with minimal intra-piece analysis, highlighting d'Indy's primary focus on Beethoven's piano sonatas.

2.4 Terminology for groupings of different structural levels

In d'Indy's chapter, some terms are defined in relation to each other, suggesting that some of them constitute a hierarchical structure. As a general guideline, a musical idea is equivalent to a theme, as seen in the main theme and subordinate themes, and is composed of phrases. Phrases are further broken down into periods, each of which is made up of cells, also referred to as germs or monads.³

²This temporal division was first proposed by Wilhelm von Lenz [Lenz, 1909] and is still in use today, although the exact divisions may vary slightly. For example, Maynard Solomon categorizes Beethoven's compositions into four main periods: "Bonn," "Vienna: Early Years," "The Heroic Period," and "The Final Phase" [Solomon, 1998].

³However, there are instances where this hierarchical structure does not hold, some of which are paradoxically noted by d'Indy himself in his schema, and others that emerge during the analysis, discussed later in this study.





Figure 2.1: "Hammerklavier", mov. 1., *Seconde idée*, first phrase (b'), mm. 63-74 [d'Indy, 1909, p. 269].



Figure 2.2: Materials beginning the second phrase (b''), mm. 75-99 [d'Indy, 1909, p. 270].



Figure 2.3: Materials beginning the third phrase (b'''), mm. 100-123 [d'Indy, 1909, p. 271].

In his provided and clearly informed analyses, the *Seconde idée* of the first movement of "Hammerklavier" could be considered as a standard example encompassing all four levels of structural units to be discussed in this section: cell, period, phrase, and musical idea [d'Indy, 1909, pp. 268-73]. He identifies measures 64-126 as representing the *Seconde idée*, labeled as "B" (therefore the Subordinate Theme, presented in the Subordinate key, here, G major), in contrast to the *Première idée*, marked "A" (or Main Theme), from earlier in the movement.

In musical idea B, three phrases are found, labeled as (b'), (b''), and (b'''), in measures 63-74, 75-99, 100-123 respectively. This illustrates a typical structural feature of Beethoven's sonata movements, as d'Indy observes through a formal labeling of all 32 sonatas, where, with a few exceptions, the *Seconde idée* consistently contains three phrases in a sonata movement [d'Indy, 1909, pp. 327-69].

Further, he subdivides the first phrase into three periods: *Première Periode*, *Seconde Periode*, and *Troisième Periode*, in measures 63-66, 67-70 and 71-74 respectively, as shown in Fig. 2.1.

D'Indy notes that the *Première Periode* itself is composed of two clearly defined cells, labeled [a"] and [b'] , as shown in the first system of Fig. 2.1.



After briefly outlining the theme to locate its elements, we will now proceed to define each part in detail, examining the structure step by step from smaller units to larger ones, following d'Indy's order of *Musical Idea Genesis*, where small units generate larger units:

Logically, it is the motif that should generate the theme, or, to use the terminology previously adopted, it is the cell that generates the period
 [...] The function of this period is indeed to generate the phrase [...]
 [d'Indy, 1909, p.239, *translation mine*]

As a result, *genesis*, according to d'Indy, can be understood as a process where

lower-level sub-units combine to form larger structures. This hierarchy of generation, moving from the motif to the period and beyond, reveals d'Indy's attempt to establish a systematic approach that ties musical form to the growth of its smallest components.

2.4.1 Cell

 A cell (cellule), also referred to as a "germ" (Germe) or "monade," is an indivisible element in the musical hierarchy [d'Indy, 1909, p. 234]. Although a cell and a motif are not identical for d'Indy, a cell aligns with a motif; in other words, the cell serves as the container, while the motif is the content within that container. D'Indy writes,

Just as the living cell contains the germ, so the musical cell contains, in a way, the motif. In both cases, the container is indissolubly linked to the content: neither can be conceived of separately. There cannot be a cell without a motif, musically speaking, and the decomposition of either, or even their separation under the pretext of analysis, would rightly appear unintelligible, if not unintelligent. [d'Indy, 1909, p.234, *translation mine*]

Cells can be categorized as rhythmic, melodic, or harmonic, with each type being dominated by its respective aspect: rhythm, melody, or harmony. It is important to note, however, that this classification does not exclude other qualities. For instance, a melodic cell is predominantly melodic in nature, but it can also exhibit minimal rhythmic or harmonic dimensions. For example, in Beethoven's "Hammerklavier" shown in Fig. 2.4, the first two cells are identified as rhythmic, while the last is classified as melodic, despite their shared solid-chord texture, which will be further dissected later. Thus, this taxonomy emerges from an adequate generalization—helpful for identifying primary characteristics, but not rigidly definitive.

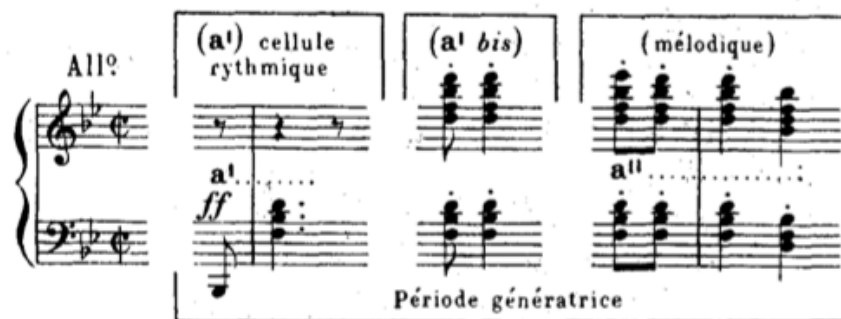


Figure 2.4: Rhythmic and Melodic Cells found in Beethoven's "Hammerklavier" [d'Indy, 1909, p. 236].

Rhythmic Cell The opening two measures of Beethoven's 5th Symphony, shown in Fig. 2.5, exemplify rhythmic cells, where the motive consists of a eighth-note rest, 3 eighth-notes followed by a longer note, displaying pronounced rhythmic characteristics.



Figure 2.5: mm. 1-5, a rhythmic cell with its repetition, followed by a generative period [d'Indy, 1909, p. 236].

Pitch-wise, this motive consists of only two tones that leap to a third, resulting in a diminished melodic character and insufficient tones to clearly define a harmony, which highlights the rhythmic characteristics even more prominently. As a result, downplaying the role of the other two dimensions further emphasizes the primary rhythmic aspect.

Another example of a rhythmic cell builds on the earlier discussion of the "Hammerklavier" example shown in Fig. 2.4, where the motif is an anacrusic gesture: an eighth note on the weak part of a weak beat, followed by a longer note on a strong beat. Pitch-wise, a significant leap creates a non-cantabile thus non-melodic quality. Additionally, the cell itself does not fully represent the harmonic region it belongs

to, as the harmony is articulated over the span of the two measures, rather than by this cell alone, thus also suggesting a minimal harmonic characteristic.

Melodic Cell The third cell of "Hammerklavier" example just discussed is identified as a melodic, in which the upper voice features a descending second, a repeated note and a descending third leap, significantly more cantabile than the previous two cells, displaying a relatively stronger melodic characteristics.

Another example of a melodic cell is found in the beginning of Beethoven's Symphony No. 6, Op. 68, where there are two step-wise motions note pairs (A4-B $\flat$ 4 and D5-C5), shown in Fig. 2.6. However, these cells do not contain purely linear motion; they include a leap of a third from B $\flat$ 4 to D5, which still remains cantabile, preventing the cells from becoming scale-like and thus making them more distinctly melodic.

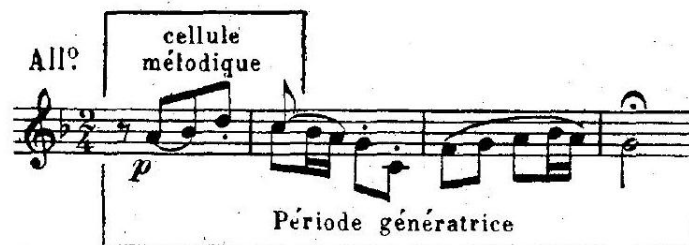


Figure 2.6: Melodic Cell found in Beethoven's Symphony No. 6 [d'Indy, 1909, p. 236].

Harmonic Cell An example of a harmonic cell is found at the beginning of Beethoven's Sonata, Op. 81a "Lebewohl," shown in Fig. 2.7, where the texture consists exclusively of diaphony with a horn-call gesture.⁴

⁴This diaphony mimicks a horn-call gesture by moving from a major 3rd (E $\flat$ to G) to a perfect

5th (D to B $\flat$), defining incomplete I and V chords.





Figure 2.7: Initial Harmonic Cell from Beethoven's Sonata, Op. 81a, No. 1 "Lebewohl" [d'Indy, 1909, p. 237].

espace vide inutile

Despite the diaphony suggesting an incomplete harmony for each two-note intervallic structure, it implies an I-V-I progression, with the middle note interpreted as a passing harmony. As a result, this complete cell can be viewed as a prolongation of the tonic harmony. Furthermore, the bass line arpeggiates all the chord tones of the tonic chord, further insinuating an overall tonic harmony. The even duration of each note, unlike the short-long alternation seen in previous examples, combined with the purely stepwise motion in the upper voice, minimizes both the melodic and rhythmic dimensions. As a result, this cell exhibits a strong harmonic character.



However, d'Indy offered an alternative labeling for that cell in a different context. For instance, in the "Histoire" section, d'Indy referred back to this cell as "cellule x" and paired it with a neighboring rhythmic cell, "cellule y" [d'Indy, 1909, p. 255]. In this analysis, he identified "cellule x" specifically as a melodic cell, noting that it generates the second idea (B) of the movement and plays a significant role in the terminal development of the section. Meanwhile, he labeled "cellule y" as a rhythmic cell that gives rise to the first idea (A).

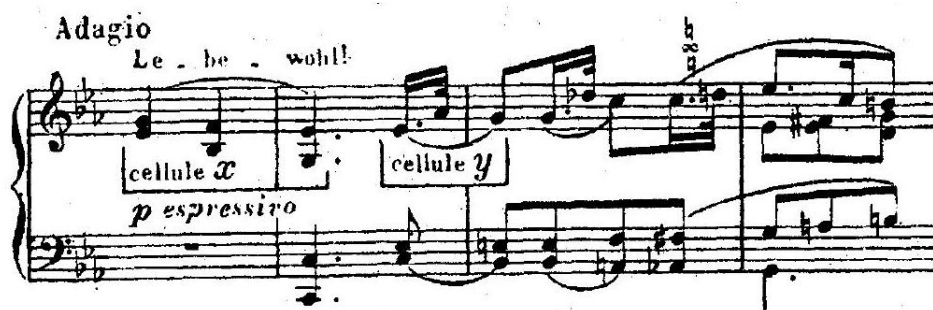


Figure 2.8: Beethoven's Sonata, op. 8, No. 1 "Lebewohl", mov. 1, mm. 1-4 [d'Indy, 1909, p. 255].

This distinction highlights d'Indy's view that cells can take on different roles, melodic or rhythmic, depending on their thematic function within the piece. On one hand, we can speculate that a cell could be categorized based on its behavior in the work. On the other hand, d'Indy demonstrates a degree of inconsistency that may seem excessive; for instance, Campos criticized d'Indy's emphasis on inter-cellular relationship as exaggerated [Campos, 2013, pp. 79-84].



Figure 2.9: Beethoven's Sonata, op. 8, No. 1 "Lebewohl", mov. 1, mm. 50-2 [d'Indy, 1909, p. 255].

Shown in Fig. 2.10, d'Indy observes that cells x and y, initially identified in the introduction of Beethoven's sonata, reappear here in a structured and purposeful way. He views cell x as a skeleton—a fundamental descending line that provides structural grounding. Along this skeleton, the rhythmic cell y appears three times, creating a recurring pattern that follows the descending trajectory established by cell x.

Campos points out that labeling the passage in Fig. 2.10 as containing 3 re-

occurrences of cell *y* raises difficulties, as it is unclear whether its identity is defined primarily by its initial rhythm (♪♪♪), its interval pattern, or its accentuation. He suggests that the size of this combination is insufficient to establish meaningful thematic connections.



Figure 2.10: Beethoven's Sonata, op. 8, No. 1 "Lebewohl", mov. 1, mm. 16-20 [d'Indy, 1909, p. 256].

However, while this claim may seem exaggerated, d'Indy's analysis consistently centers around this concept and applies it repeatedly. As we will see later in his analysis of the Hammerklavier, d'Indy links different formal sections within a single movement through the recurrence of motifs contained within cells. Furthermore, this approach to motivic connection extends beyond individual movements, as seen in Example 2.11, where cells *x* and *y* are identified in the Finale movement. This inter-movement association lays the groundwork for d'Indy's discussion of cyclic sonata in his subsequent chapter.⁵

⁵In Chapter 5 of Volume 2 - Part 1, d'Indy describes the cyclic sonata as a form in which the construction is unified by recurring thematic material, which reappears in various forms across each movement and serves a regulatory or unifying function. This cyclic quality, created by the presence of recurring themes or "motifs conducteurs," lends each movement a sense of interdependence within the larger work. While cyclic elements are not exclusive to the sonata, d'Indy considers the sonata to be the prototype for all symphonic forms that later adopted cyclic structure. He credits Beethoven with pioneering this approach, which was fully realized by César Franck [d'Indy, 1909,

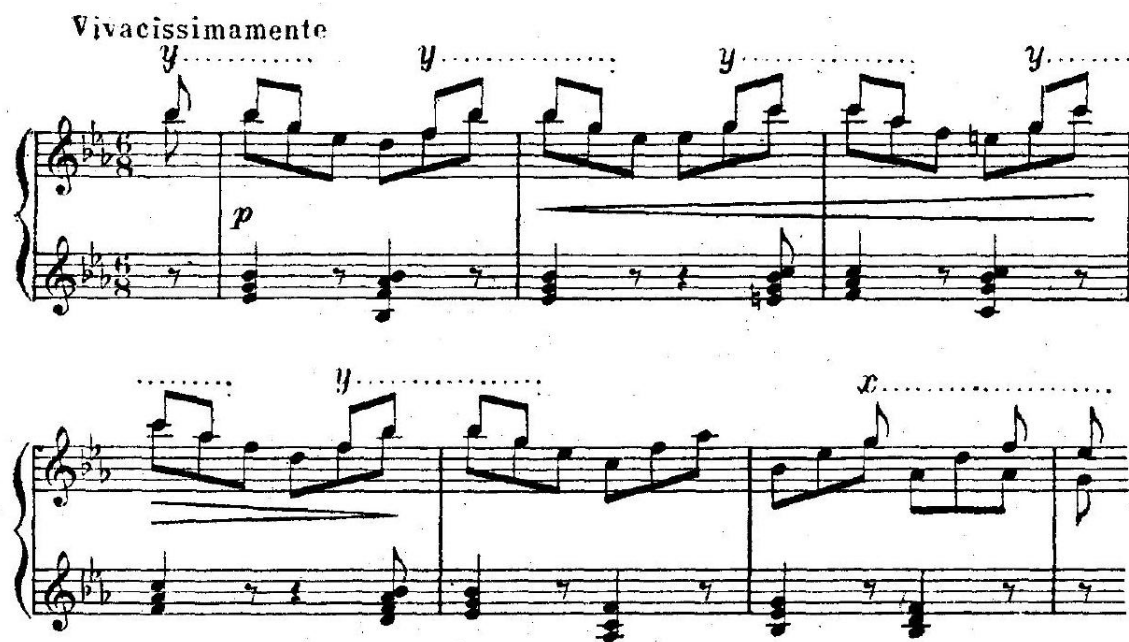


Figure 2.11: Beethoven's Sonata, op. 8, No. 1 "Lebewohl", mov. 3, mm. 10-6 [d'Indy, 1909, p. 256].

2.4.2 Period

A period, as a structural unit in d'Indy's discourse, located in between the cells and the musical ideas (themes), typically consists of a combination of multiple cells. It often receives an ordinal attribute based on its position within a larger structural unit⁶, resulting in labels such as the initial period, secondary period, and third period. The most commonly referenced type of period is the initial period, also known as the principal period, but it is most frequently referred to as the generative period [d'Indy, 1909, pp. 234-5].

Like a cell, the term period refers to a grouping entity, with each period containing specific thematic or motivic content within its boundaries. This container-content relationship applies to all periods in general. However, the generative period holds a unique role: it introduces the fundamental thematic cells, with subsequent

⁶Such larger structural unit could be either a musical idea or a phrase, which will be introduced while discussing *Phrase* as a structural unit in d'Indy's discourse.

periods drawing on this generative theme rather than introducing entirely new motives. This relationship is the basis for d'Indy's designation of this initial period as the generative period, as it serves as the source from which the thematic content of later periods is derived, or "generated". In this way, the generative period serves as a template, establishing the primary theme that subsequent periods develop rather than introducing new motives.

The distinction between cells and periods is based on different criteria: cells are defined by the number of notes they contain, while periods are defined by the number of measures they span. A cell typically consists of a small set of notes, as it cannot be meaningfully divided beyond its constituent pitches. In contrast, a period must extend for at least two measures, as shown in Examples 2.5 and 2.4, highlighting that its length is determined by its measure count rather than the number of individual notes. Thus, the structure of a cell is constrained by its note content, while the structure of a period is determined by its rhythmic span.

Generative Period A generative period is a specific subclass of a period, as discussed above, and coincides with a generative theme. In this relationship, the same container-content analogy applies: the generative period serves as the container, while the generative theme represents the content within the region bounded by this period, as stated by d'Indy,

This principal period contains the generating theme, just as the cell contains the motif. Here again, the container is necessarily linked to the content: there is no principal period without a generating theme.
[d'Indy, 1909, p.235, *translation mine*]

Although both terms—generative period and generative theme—describe different entities, they are intertwined within the same temporal span. However, the word “generative,” which is etymologically linked to *genesis*, is more prominently associated with the period level than with other entities throughout the text, raising the question of why d'Indy privileges this term. D'Indy himself offers some insight:



Logically, it is the motif that should generate the theme, or, using the terminology we previously adopted, it is the cell that generates the period; but this latent process occurs **unconsciously** in most cases, which is why we call it the generating period when it generally presents itself to the composer's mind as a whole, rather than through successive fragments, except when it later undergoes profound voluntary and conscious modifications.

The function of this period is, in fact, to generate the phrase, or phrases, that express the idea; and this second operation is never the result of chance, pure instinct, or what is commonly referred to as 'inspiration.'

[d'Indy, 1909, p.239, *translation mine*]

Thus, d'Indy seems to privilege the term generative period because he believes that it is at this level that the composer becomes consciously aware of the material. This level marks the transition from unconscious development to deliberate compositional control, as the composer begins to organize the musical material with clear intent. The generative period, therefore, represents a turning point where creative processes become consciously directed and purposeful, explaining why d'Indy emphasizes it over other subunits.

This perspective is exemplified in his analysis of Example 2.12, where he states,

The initial cell (a') is predominantly rhythmic and tonal: it acts upon the generating period, stated twice on two different degrees, and gives this first idea a primarily masculine character.

The eighth-note pattern (a''), however, takes on a more melodic aspect in the secondary or complementary periods. This contrast does not undermine the stylistic unity of the idea, as this pattern (a''), common to all the periods, establishes a real connection between them. [d'Indy, 1909, p.264, *translation mine*]

a'' seems to keep a rhythmic dimension in addition to its melodic identity. As a



Figure 2.12: D'Indy's analysis on the opening measures [d'Indy, 1909, p. 264].

result, d'Indy views the beginning of the second period's material as originating from the a'' melodic cell, albeit with some modifications: a shift from staccato to legato, slight alterations at the start of the melodic line, yet maintaining the descending contour in the last two notes. This ending gesture reinforces the resemblance, even though the third note is elongated to a half-note, subtly preserving the underlying connection to the generative period despite the surface-level changes.

Upon investigating the generative period, the role of cells seems to come to the forefront. On one hand, d'Indy believes that at the cellular level, much of the composer's creative activity is guided by an instinctual, unconscious force. On the other hand, without subdividing periods into smaller cells, the connection between different sections might not be readily apparent—especially through a visual reading of the score rather than an aural experience. This subdivision helps to highlight how seemingly unrelated material can be traced back to earlier sections, as otherwise, such associations could remain obscured by surface-level elements, like dense chords written in succession. Therefore, breaking the music down into cells serves an analytical purpose: to reveal and articulate relationships that might remain acoustically implicit, ensuring that connections between sections are made more explicit

and justifiable from a theoretical standpoint.

2.4.3 Phrase

D'Indy provided an explicit definition of a phrase in relation to a period: a phrase is formed by the sequential arrangement of the main or generative period followed by the secondary periods, which together articulate a larger structural unit such as a musical idea or the transition. As seen in Example 2.1, this phrase consists of three periods.

However, d'Indy's assertion of "Phrase" in relation to musical idea is somewhat contradictory, as he provides two different perspectives in the same discussion. Initially, he states that a phrase encapsulates a musical idea, similar to how a principal period contains a generative theme, or how a cell contains a motif. However, he later notes that a musical idea might require multiple phrases for its complete expression, with one phrase serving as the principal one that holds the essential part of the idea. This suggests that, while a phrase can be seen as a container for an idea, a single idea may also span several phrases, thereby challenging the clarity of his terminology. D'Indy mentions,

The phrase contains the idea, just as the main period contains the generating theme and the cell contains the motif; however, although the motif is always entirely included in the cell, and the theme in the period, the musical idea is not necessarily stated by a single phrase. The full expression of the idea sometimes requires, on the contrary, two and even three phrases, one of which, the main one, contains the essential part of the idea. [d'Indy, 1909, p.235, *translation mine*]

Thus, d'Indy initially asserts that a phrase contains a musical idea, similar to how a motif is contained within a cell. But by acknowledging that a musical idea may require more than one phrase for its full expression, d'Indy creates a paradox within his own definitions—a contradiction that I will attempt to clarify after examining his concept of the musical idea.

2.4.4 Musical Idea

Musical Idea: Generalized vs. Constrained To resolve the ambiguity and avoid confusion, I propose a distinction between a generalized musical idea and a constrained musical idea, as illustrated in Table 2.5. A **Constrained Musical Idea** refers to the specific content of a single phrase container, which includes significant thematic material, typically aligning with the generative theme. On the other hand, a **Generalized Musical Idea** represents a broader formal section, equivalent to a musical theme, such as a primary theme or a subordinate theme in Caplinian sense. This ambiguity could be an example of phenomenon called "pars pro toto," meaning "a part (taken) for the whole," where a smaller component is often used to designate a larger structure containing it [Oxford Living Dictionaries, 2016].

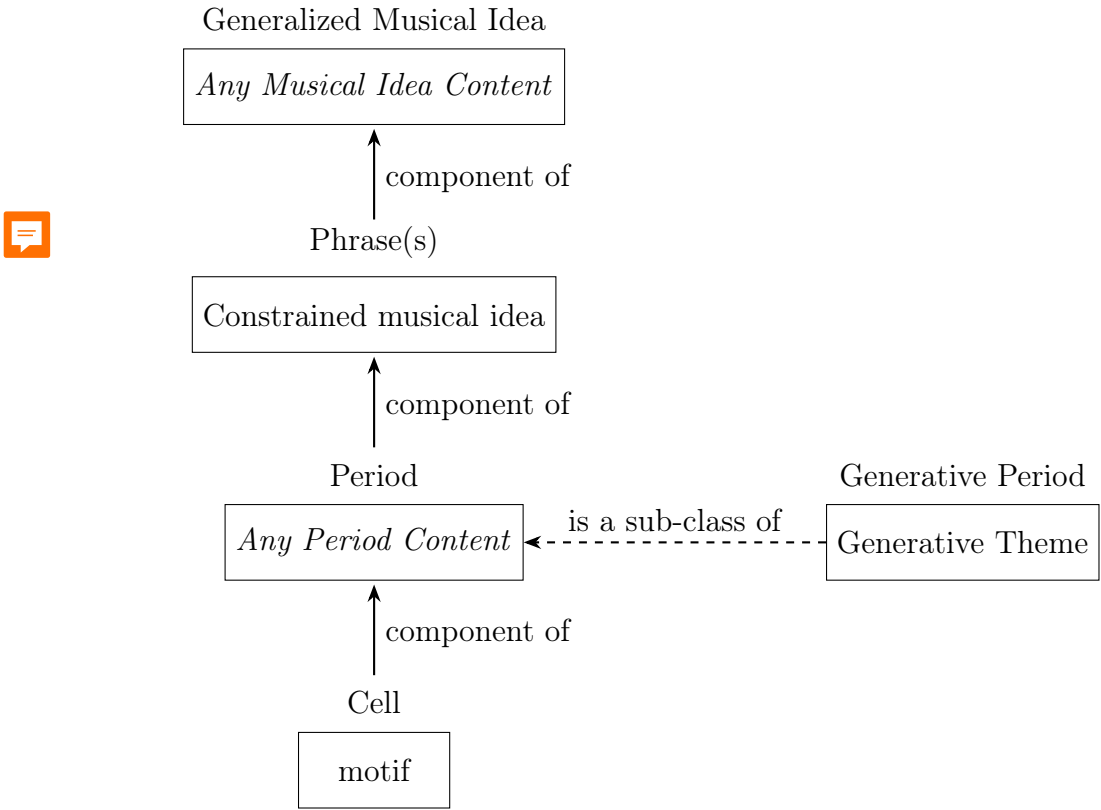


Table 2.5: An Illustration of Hierarchy of Terminologies as a rule of thumb

Therefore, d'Indy equates a single phrase with a single theme, and a group of multiple phrases with a theme group. Examples 2.1, 2.2, and 2.3, each corresponding to a phrase, collectively form a Generalized Musical Idea. Among the phrases in

this theme group, however, the first phrase in Example 2.1 is the most thematic and prominent, thus corresponding to a Constrained Musical Idea.

Genesis vs. Development of Musical Idea Genesis refers to the process of forming a musical idea, completing the initial phase during which the idea does not yet exist in its full form. Therefore, activities of genesis can be observed at formal levels below a thematic unit, such as a generative period. If the musical idea has not yet been fully established, then while materials, such as motivic elements, may appear to "develop," they still fall within the scope of genesis.

Development, on the other hand, is a technique that involves expanding and elaborating on an already established musical idea. Development extends the original idea and transforms it through variation, transposition, integration within new keys, and other techniques, allowing the idea to evolve within the musical structure and create a sense of direction and effects of surprise and reorientation.

The development technique typically occurs in two scenarios: first, after the exposition ended where both musical ideas have been introduced, such as in the development section; second, when only the primary idea has been stated, with development techniques leading into the rest, as seen in moments where materials from the first idea reappear in the transition and the second idea. Compared to the some discussion of genesis, this study barely covers development techniques, as neither of the two typical scenarios for development falls within its scope.

This study's discussion is confined to the exposition and does not extend beyond it. Consequently, the first scenario falls outside the scope of this analysis.⁷

For the other scenario, since the structural architecture above the level of the musical idea remains consistent across various theories—including d'Indy's—as nearly

⁷The development section is a formal section with less conventionality, making it less adequate to generalize than other sections. Furthermore, d'Indy's analysis of the development section is illustrated by the Hammerklavier Sonata, where the development consists largely of a fugal passage. This approach is unconventional and not fully representative of Beethoven's typical sonata developments, making it less suitable for comparative purposes in this study.

all sonata theories feature a main theme, transition, and subordinate theme in the exposition, followed by the development and recapitulation sections, reporting on d'Indy's discussion about these structural elements offers limited new insights. There is therefore significantly less discussion of development as a technique compared to that of genesis in this study, as the primary focus of this study is on structures below the level of the musical idea, whereas development as a technique pertains to activities above that level.

2.5 D'Indy's Values

Understanding d'Indy's analytical values requires examining the broader ideological and methodological principles that underpin his approach. His theories reflect a preference for homogeneity and dualistic structures, interpreting musical contrasts through a lens of binary oppositions such as gender and "masculine" vs. "feminine" themes. This tendency extends to his use of numbers and proportional systems, which impose a sense of structural order. However, these values are not purely formalistic—they are intertwined with cultural and ideological beliefs, including his documented views on race and nationalism, which permeate his veneration of certain composers and his hierarchical treatment of musical styles. To contextualize d'Indy's values, it is crucial to compare them with those of other pre-modern theorists, whose frameworks similarly blend musical analysis with broader cultural ideologies. Through this comparison, we can extract representative values that illuminate not just d'Indy's theoretical approach but also the larger cultural narratives that shaped early music theory.

2.5.1 Gendering Themes

Vincent d'Indy's exploration of sonata form places a notable emphasis on the gendering of thematic material, reflecting both the formal principles of the music and the cultural ideologies. He perceives the interaction between the primary and

secondary themes not merely as a technical contrast but as a reflection of archetypal gender dynamics. For d'Indy, these themes take on masculine and feminine characteristics, embodying broader narratives of strength, clarity, grace, and fluidity. He writes,

Strength and energy, conciseness and clarity: these are almost invariably the masculine traits of the first idea: it imposes itself in vigorous and abrupt rhythms, clearly affirming its tonal property, singular and definitive.

The second idea, on the other hand, full of sweetness and melodic grace, almost always displays through its prolixity and modulating indetermination traits that are eminently feminine: supple and elegant, it gradually unfolds the curve of its adorned melody. [d'Indy, 1909, p.262, *translation mine*]

Through this lens, d'Indy not only reinforces the established understanding of thematic contrast but also extends it to an almost symbolic level, where the primary theme's firm tonal grounding and energetic rhythmic structure represent masculine authority. The secondary theme, on the other hand, is imbued with a feminine quality, characterized by its melodic intricacy and harmonic instability.

This gendered symbolism resonates with the earlier ideas of theorists like A.B. Marx. d'Indy posits a duality of male and female principles in music, reflecting a broader cultural tendency to ascribe gender roles to abstract concepts. Marx had previously articulated a similar duality in musical phrases, associating strong, directive gestures with masculinity and more passive, resolving gestures with femininity [Burnham, 1996].

In this sense, d'Indy and Marx are part of the same theoretical lineage that seeks to explain musical drama as the interaction between opposing forces—forces that must ultimately be reconciled in the form's resolution. While the gendered metaphor is a striking aspect of d'Indy's writing, it is only one manifestation of a much larger, more abstract concept of duality that pervades his thinking on music.

2.5.2 Dualism

Marx's application of dualism to various musical elements does not limit to gender analogy, but he also proposed, such as *Gang* and *Satz*—terms that describe the forward motion of music (*Gang*) versus its state of rest or closure (*Satz*) [Marx, 1879].

Similarly, d'Indy introduces his own dualistic terminology, referring to the *état de marche* and *état de repos*, also known as *état de translation* and *état d'immobilité* respectively [d'Indy, 1909, pp. 243-4]. The *état de marche*, akin to Marx's *Gang*, is characterized by a sense of movement, drive, and progression, which we can easily associate with the "masculine" principal theme. The *état de repos*, much like Marx's *Satz*, is a state of rest or resolution, mirroring the qualities of the "feminine" secondary theme, which is more expansive, lyrical, and modulatory. This structural dualism not only reflects d'Indy's preference for contrasting forces but also reinforces the broader 19th-century intellectual tradition of viewing musical form through the lens of oppositional pairs, whether gendered, dynamic, or harmonic.

2.5.3 Number Favour

Beyond duality, Vincent d'Indy's theoretical favour on the number three as a structural and generative element permeates his music analysis. This manifests in various aspects of his theoretical system, such as the three types of cells and developments (rhythmic, melodic, and harmonic), and the three types of modulation (passing, accidental, definitive). A tripartite structures also surface in his formal subdivision, where in his second book, he often identifies three phrases in the second idea section such that section of "Hammerklavier" discussed earlier, and three periods in a phrase, such as in 2.1. There is therefore a tripartite structure nested inside another tripartite structure.

This numerical emphasis parallels Heinrich Schenker's privileging of the number five, which he views as a synthesizer of musical coherence [Clark, 1999]. As described by Clark, according to Schenker, all musical elements are generated from a single

origin through organic replication, and the number five could be traced in various places. Schenker's theory of the fundamental line derives from the notion that the number five is the essential structure underlying all tonal music, a view that contrasts but runs alongside d'Indy's focus on the number three.

Number plays a crucial role in understanding the concept of generation: by using a consistent numerical basis, it ensures internal coherence, uniformity, and a clear hierarchical structure, which collectively form the foundation of organicity. With one consistent rule applied throughout, the structure replicates itself naturally, creating a sense of growth and development akin to an organism expanding according to its inherent logic.

2.5.4 Generation

The concept of generative structures further links d'Indy's theoretical approach to that of Schenker, and the use of number contributes to the process of generation. Prior to both scholars, the idea of generation in music theory was confined primarily to the realm of harmony. Introduced by Rameau, this concept was understood as the generation of chord tones from the *corps sonore* or fundamental tone [Rameau, 1966].

However, Rameau's generative approach relies heavily on assertions, supported only as far as natural phenomena can explain them. Once this explanatory power breaks down—such as in the case of generating a minor chord—his theory collapses. For instance, while the overtone series can justify the generation of a major chord, it cannot logically extend to the minor chord, forcing Rameau to treat it as a mirror image rather than a true product of acoustic laws. To some extent, relationships in generative processes are established through assertions, though they may be supported by analogical or epistemological interpretations. Nevertheless, these interpretations remain secondary, as the primary focus is on maintaining internal consistency.

By the time of d'Indy and Schenker, the notion of generation had expanded be-

yond harmony to encompass formal structures, with both theorists striving for a more uniform and hierarchical system. Numbers are central to their frameworks, providing a consistent basis for structural organization. Schenker's "five generates all" principle uses the descent from the fifth scale degree to shape the tonal and harmonic framework of an entire composition, suggesting an organic unity throughout. Similarly, d'Indy applies a generative concept using a "three-period phrase" as a foundational building block from which larger forms are constructed. This use of consistent numbers imbues their theories of organic replication or *Genesis of Musical Idea*, reinforcing the perception that each formal unit is a natural outgrowth of its predecessors.

In summary, d'Indy's concept of generation, by referencing theories developed from other theorists, can be understood as the association of formal subunits through a consistent numerical framework. However, the internal content of the generative source also plays a crucial role in shaping the generation process, as later elements are seen as organic replications or "Geniture" of earlier material.⁸ For instance, Rameau's generative source, the *corps sonore*, uses frequency as its defining attribute, with numerical ratios establishing harmonic relationships based on that attribute. In contrast, for Schenker and d'Indy, who operate at the level of form, a wider array of attributes is available. Motivic content becomes a key element, prompting them to a strong emphasis on homogeneity in their analyses. This emphasis further supports their view that subsequent materials are logical extensions or transformations of a primary generative source, reinforcing the perception of organic development throughout the musical structure.

Generative theory of tonal music Later scholars Fred Lerdahl and Ray Jackendoff introduced the Generative Theory of Tonal Music (GTTM) [Lerdahl and Jackendoff, 1983], which bears a terminological resemblance to d'Indy's concept of generation. GTTM, derived from Chomsky's generative grammar, is structured as a recursive, tree-like model that proceeds from top to bottom, breaking down music into hierarchical lay-

⁸"Geniture" is referring the result of a *Genesis*.

ers. GTTM comprises four primary structures: Grouping Structure, Metrical Structure, Time-span Reduction, and Prolongational Reduction. Of these, Time-Span Reduction focuses on local harmonic relationships, closure, and cadential points within smaller segments, while Prolongational Reduction addresses global harmonic continuity, organizing the music's overarching tonal direction and stability.

In d'Indy's analysis of Beethoven, Metrical Structure—the systematic layering of strong and weak beats—is not a primary focus. Instead, his emphasis is on the grouping and development of thematic and motivic elements, aligning more closely with GTTM's Grouping Structure. In his framework, individual cells are grouped together to form periods, multiple periods are grouped into phrases, and multiple phrases combine into larger sections such as musical ideas. 

While both d'Indy and GTTM use the term generation, they apply this term to designate different processes. For d'Indy, generation means building from the atomic level, where fundamental units or cells progressively group together into periods, phrases, and eventually larger formal structures. In contrast, GTTM understands generation as a deductive, grammar-like process, where overarching structural principles generate the finer details of the music, guiding each level of hierarchy from the top down.

Though both approaches use the term "generation" and applied it at the formal level—unlike Rameau's use of similar terminology applied in harmonic level—d'Indy's method remains fundamentally distinct and original, by constructing form by aggregating small units, whereas GTTM derives detail from broad structural rules. Upon noting the terminological resemblance between d'Indy's concept of generation and that of others, it becomes clear that his application of the term is quite remarkable and distinct from similar theoretical uses.

2.5.5 Homogeneity, Parallelism and Motivic Coherence

The dualistic mindset of d'Indy also leads d'Indy to emphasize homogeneity within each side of the binary opposition. Furthermore, his tendency to work with

a tripartite structure leads him to cluster materials based on shared characteristics. As a result, materials with a common texture tend to be grouped together as part of the same category, reinforcing internal cohesion within each side of the binary opposition. This approach aligns with the parallelism principle of GTTM discussed above, where similar patterns are grouped together to create perceptual coherence. This emphasis on uniformity can sometimes lead to interpretative issues, particularly when it causes him to overlook conventional formal boundaries. A striking example of this occurs in his analysis of the first movement of Beethoven's Moonlight Sonata. Rather than treating it as a sonata-form movement, d'Indy labels it as a "Lent" type piece [d'Indy, 1909, pp. 343].

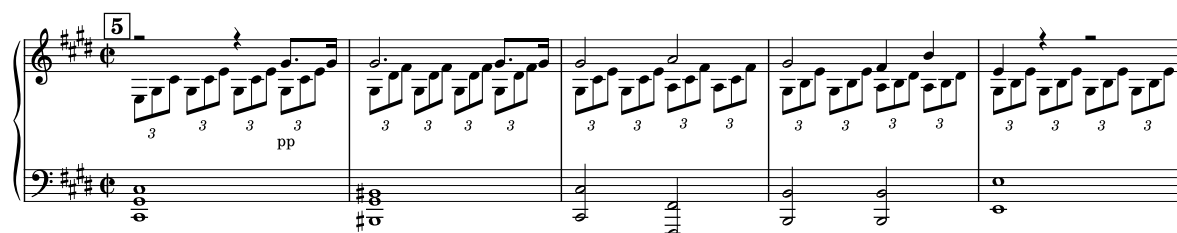


Figure 2.13: "Moonlight" Mov. 1, Main Theme.



Figure 2.14: "Moonlight" Mov. 1, Second Theme.

The movement's uniform triplet texture, which remains prominent throughout, may obscure its formal sections, leading d'Indy to prioritize its surface homogeneity over its structural logic. This unconventional approach complicates comparative analysis, as his labels diverge from contemporary formal interpretations, thereby creating discrepancies in understanding the movement's structure. This tension between duality and homogeneity in d'Indy's analytical method becomes a key factor in the contrasting formal interpretations explored in subsequent chapters.

Besides, Schenker also values homogeneity, as his idea of the *Linkage technique*,

which he introduced in his analysis of Brahms's "Variations and Fugue on a Theme by Handel, op. 24," from "Der Tonwille 8/9" ([1924]), is essentially about associating materials that are homogeneous [Baker, 2020, p. 101]. This technique emphasizes the repetition and imitation of motives, incorporating variations and contrasts rather than exact duplications.

At its core, the *Linkage technique* enhances the music's internal coherence and integrity by connecting melodic, harmonic, and rhythmic elements. This approach transcends simple thematic repetition by showcasing the extension and transformation of motives across different dimensions, thus revealing deeper structural relationships beneath the surface of the music.

2.5.6 Racism

Generalization is often necessary when constructing theoretical frameworks, as it helps simplify complex ideas into manageable categories. Favoring certain choices—such as number—stands out because of its quantifiable nature, lending itself to clear, directly measurable distinctions that can structure abstract concepts more concretely.

However, generalization can easily slip into stereotypical or reductive thinking. An overreliance on drawing parallels to nature, or claiming "natural laws" as justifications, often leads to oversimplification and a stereotyped worldview. This is particularly problematic when applied beyond music theory, as seen in the dangerous applications of social Darwinism. D'Indy and Schenker, both known for their organicist approaches to form analysis, are prone to similar pitfalls when they impose analogies from natural phenomena onto their theoretical models, risking a reductionist perspective that echoes these broader misapplications.

Both d'Indy and Schenker entwine their music theories with overt ideological positions, particularly nationalism and racism. D'Indy's antisemitic views documented by [Pasler, 2007] and [de Médicis, 2024], is his veneration of Beethoven as a symbol of musical and cultural purity. Like d'Indy, Schenker favored German composers, elevating their works as exemplars of his theoretical system. However,

he makes an exception for Frédéric Chopin, whom he refers to as an "honorary German," acknowledging Chopin's genius but placing it within a framework that reinforces the superiority of Germanic musical traditions. Schenker's similar nationalistic leanings align with his ideological belief in the supremacy of German music, particularly that of the Austro-German tradition, while dismissing non-Germanic influences. Both theorists construct their music theories to support these broader ideological views, intertwining their aesthetic judgments with cultural and racial biases.

2.6 Conclusion

In conclusion, d'Indy's approach aligns with earlier stylistic conventions that contemporary perspectives now challenge. Taking his example of gendering themes, today's understanding of gender and identity rejects the strict binary model he employs. The notion that contrasting musical forces must inherently embody "masculine" or "feminine" traits is now seen as overly simplistic and reductive. Modern perspectives emphasize that identity and expression exist on a spectrum, moving beyond a rigid "either/or" framework toward a more inclusive view that allows for "both/and" or even non-binary interpretations. This shift challenges the very premise of d'Indy's gendered oppositions, highlighting that his approach is not just outdated but also deeply rooted in cultural biases that shaped the broader ideological narratives of his time.

In contrast, contemporary theories, such as William Caplin's theory of formal functions, are largely free of overt ideological biases. Caplin acknowledges the historical use of gendered themes by theorists like A.B. Marx in his textbook, noting it as a historical fact but refraining from incorporating it into his own theoretical framework [Caplin, 2013, p. 358]. Caplin's approach is more concerned with providing a neutral, systematic analysis of musical form, emphasizing practical and structural observations without layering them with cultural or ideological interpretations. His work engages with music on its own terms, offering clear examples and analytical

categories that prioritize the internal logic of the music over symbolic narratives.



Comparing d'Indy's theory with Caplin's is valuable because it underscores a fundamental shift in music theory: from interpretive models laden with ideological assumptions to frameworks that aim for a more objective, context-independent analysis. This contrast not only reveals the biases in earlier theories but also illustrates how contemporary theory seeks to disentangle analytical methods from subjective or culturally contingent narratives. Understanding these differences is crucial in evaluating how music theories have evolved and the ways in which modern methodologies strive for analytical clarity and inclusivity.

2.7 Caplinian Theory Overview

This study presumes a certain level of familiarity with Caplinian analysis, as it is widely recognized within the field of music theory.

William Caplin first introduced the concept of "Expanded Cadential Progressions" in his 1987 article [Caplin, 1987]. He expanded on these ideas in his 1998 textbook, *Classical Form: A Theory of Formal Functions for the Instrumental Music of Haydn, Mozart, and Beethoven* [Caplin, 1998]. In his 2000 article, "Harmonic Variants of the 'Expanded Cadential Progression,'" Caplin refined his earlier concept by proposing that certain harmonies within these cadences could be substituted with chromatically altered chords [Caplin, 2000]. His 2004 article, "The Classical Cadence: Conceptions and Misconceptions," reaffirmed his views on cadences, highlighting the differences between his approach and traditional harmonic theory, and offering insights into common misunderstandings about cadences [Caplin, 2004]. Caplin further developed these ideas in his 2013 textbook, *Analyzing Classical Form: An Approach for the Classroom* [Caplin, 2013], and in his 2018 article, "Beyond the Classical Cadence: Thematic Closure in Early Romantic Music," where he applied his theoretical concepts to the analysis of Romantic-era compositions [Caplin, 2018].

Caplin's most significant theoretical contribution is his framework of formal functions, which categorizes a musical work or section into distinct stages such as be-

ginning, middle, and ending, with these stages represented through harmonic progressions identified as "prolongational," "sequential," or "cadential," respectively [Caplin, 2013, p. 35]. This approach applies to larger structures like the exposition, development, and recapitulation, as well as smaller intra-thematic structures, such as the presentation, continuation, and cadential phrases in a sentence theme type. Furthermore, Caplin's theory examines whether a section is "tight-knit" or "loose" in terms of its phrase-structural organization, noting that a sentence is a looser theme type compared to a period theme type [Caplin, 2013, p. 203-4].

Caplin's explains the "sentence" theme type as a conventional yet looser theme structure within classical composition. The sentence begins with a presentation phase that typically prolongs the tonic harmony, setting a stable foundation. This is followed by a continuation phase, which may use one or several of various techniques such as fragmentation and an acceleration of both surface and harmonic rhythm, marking a clear middle section that often includes sequential progressions. The end of the sentence is articulated by a cadential idea in a cadential phase, which is syntactical and signifies a conclusive end to the thematic unit. This structure allows Caplin to demonstrate how thematic development and formal functions interact within the broader context of classical music's phrase structure.

Caplin's explains the "period" theme type as a tight-knit and more balanced structure compared to the sentence [Caplin, 2013, p. 203-4]. A period consists of two main phrases: the antecedent and the consequent, each typically spanning four measures, contributing to its symmetrical form. The antecedent phrase ends with a weaker cadence—most often a half cadence or an imperfect authentic cadence—creating a sense of incompleteness that prompts a continuation. This cadence establishes a point of tension or pause, setting up the expectation for a stronger resolution. The consequent phrase then mirrors the antecedent's basic melodic material, but with variations that lead to a more conclusive cadence, typically an authentic cadence [Caplin, 2013, p. 73-85]. This closing cadence provides a symmetry to the period and syntactical closure, effectively reinforcing the tonal goal and



unifying the phrase structure. 

It is essential to acknowledge that Caplin's theory, as a contemporary analytical framework, does not exist in isolation. It frequently incorporates and references other theoretical frameworks, including ideas from Schmalfeldt, Hepokoski, and Darcy. For example, in *Analyzing Classical Form*, terms such as "One More Time" and "Medial Caesura" are integrated into the analysis, highlighting the dialogue between Caplin's ideas and existing theories [Caplin, 2013, pp. 143, 310].

Moreover, Caplin's theory is so influential that many scholars have created their own versions based on his ideas. These variants, that I will refer later in the text as "Neo-Caplinian," share a common theoretical ground. They build around Caplin's concepts and terminology, meaning they are part of the same analytical discourse, even if each scholar brings their own perspective.

Therefore, it is more accurate to describe the upcoming analysis not as a confrontation of Caplin's and d'Indy's theory, but rather of contemporary theory and those articulated by d'Indy's, whose ideas also resonate with the broader theoretical landscape of his time, as evidenced by the resemblances identified earlier. Therefore, the interaction between d'Indy's and Caplin's theories can be seen as a dialogue between two epochs. Such a comparative analysis could yield valuable insights that contribute significantly to the field of *New Formenlehre*.

Part II

Analysis